



Adaptive FEM-BPNN model for predicting underground cable temperature considering varied soil composition

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ABSTRACT

In underground cables of power systems, the maximum temperature of the cable is a crucial factor in determining its capacity. According to standards, the permissible operating temperature for the XLPE cable conductor under steady-state conditions is 90 °C – a limit that should not be exceeded. Exceeding this temperature may lead to a thermal breakdown in the cable insulation, thereby resulting in interruption of the electrical power supply. Many factors affect the cable temperature, particularly through the processes of heat dissipation and diffusion from the cable into its surroundings. These factors include soil types and compositions, cable installation configuration, and thermo physical properties; therefore, accurate analysis of these factors is crucial for cable loading. In this study, the finite element method (FEM) is employed to predict the cable temperature considering different soil compositions and to present a new approach for the thermal analysis of an underground cable system. The novel approach considers various environmental conditions including single-layer and multi-layer soil types, homogeneous and non-homogeneous soil compositions, two configuration types – flat and trefoil – as well as two types of backfill materials, specifically sand-cement mixture backfill (SCMB) and fluidized thermal backfill (FTB), and dry zones to offer deeper insight into a thermal analysis. Given that the FEM requires the construction of a complex geometric model within an optimal operating condition to obtain results with high accuracy—a process that can often be complex as well as not adaptable because it depends on constant mathematical calculation—This paper presents a novel approach FEM-BPNN that uses an adaptive Backpropagation neural networks (BPNN) model as its mainstay. The proposed BPNN model exploits historical data from FEM to refine its predictive power, therefore, increasing its efficiency and accuracy. Furthermore, the model is subject to an optimization process, adjusting and refining its internal parameters in response to new data, with the ultimate goal of improving the predictive model capabilities for the temperature of underground power cables. The results underscored the high performance of FEM in the simulation, and it was observed that FEM yielded results closely aligned with those of the IEC standard. Moreover, the proposed FEM-BPNN demonstrated exceptional accuracy, achieving a low RMSE score of 0.008. It also exhibited impressive performance in the linear regression analysis, with an R^2 value of 0.99. These metrics collectively signify the robustness and efficacy of the model.

1. Introduction

Heat transfer is an essential aspect of many engineering applications as it impacts the performance and safety of various devices, systems,

main components [1–5]. In the context of underground cables, cable capacity is an essential parameter that determines the maximum rated current that a cable can safely carry without exceeding its temperature limits [6–9]. In a non-optimal environment, the temperature of the cable

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core can exceed its designed operating thresholds, leading to insulation degradation, cable damage, and even electrical fires. On the other hand, heat transfer analysis focuses on the flow of thermal energy due to temperature differences of the cable materials. Finite element method (FEM) can be used to simulate heat transfer in underground cables and predict the temperatures of the cable elements and its surrounding environment. This information can be useful in estimating the temperature rise of the cable and its effect on the cable capacity.

Effective heat transfer management is crucial for ensuring the safe and efficient operation of underground cables. The problem statement in heat transfer for underground cables is multifaceted, necessitating a systematic and scientific approach to address the various issues. The main challenge is to develop a robust and accurate method for calculating the temperature around underground cables, taking into consideration the complex interplay between various factors such as soil types and compositions, cable installation configuration, and thermo physical properties. The methods of temperature calculation are essential for engineers and can be utilities to predict and monitor cable temperatures, ensuring that cables operate within their thermal limits and prevent overheating incidents. Furthermore, the impact of cable geometry models and cable configurations, such as flat and trefoil arrangements, on heat dissipation and heat diffusion from the cable core to the surface ground must be investigated. This requires examining the influence of cable spacing, depth, and installation methods on heat transfer management and understanding the trade-offs between various cable configurations and their thermal performance. Another aspect of the problem statement involves analyzing the effects of homogenous and non-homogenous soil on heat dissipation and diffusion [10–13]. This includes understanding how variations in soil composition, moisture content, and density can influence heat transfer and cable performance. Additionally, the potential variations in soil properties due to dry zone formed-induced effects need to be considered, as they may significantly impact the thermal performance of underground cables.

Many research papers have been published on the topic of thermal performance of underground cables. Karahan et al. [14] investigated the effects of electrical parameters on cable core temperature, which in turn affects cable ampacity. The authors employed the FEM to calculate temperature distribution, utilizing an electric-thermal combined model that incorporated environmental factors and cable properties. The study revealed that factors like soil thermal resistivity and weather conditions play a role in affecting the current carrying capacity of power cables. Furthermore, the electric-thermal combined model proves effective in accurately determining both temperature distribution and cable ampacity. Al-Saud et al. [15] proposed a new algorithm for perturbed finite-element analysis to calculate thermal field sensitivities and capacity of underground power cables subject to variations in the cable thermal circuit parameters. The proposed methodology was validated through laboratory experiments and applied to various practical cable systems. Hwang and Jiang [16] proposed electromagnetic FEM for a thermal analysis of underground power cables directly buried in banks and PVC ducts. Liang [17] presented the challenge of accurately computing permissible conductor currents in underground power cable systems, which are often operated at maximum capacity. The author employed the streamline-upwind/ Petrov-galerkin (SUPG) and FEM for steady-state thermal analysis and verified it with a full-size experimental setup. The study concluded that the SUPG method offered better calculation accuracy than traditional analytical methods, particularly in accounting for convection-induced heat exchange.

Al-Saud et al. [18] presented an optimization problem focused on thermal field parameters for underground power cables. Utilizing nonlinear programming and finite element analysis, the authors employed methods like generalized reduced gradient (GRG) to solve the complex issue. They found optimized thermal conductivities for various layers around the cables, achieving temperature reductions and ampacity improvements in different scenarios. Li et al. [19] presented a method for real-time assessment of underground cable ratings using

distributed temperature sensing (DTS). Addressing the dynamic nature of soil thermal properties, the authors employed a combination of FEM and gradient-based optimization to estimate these parameters. They found that their approach accurately predicted cable surface temperatures and improved operational efficiency, validating their method both in the lab and on a 230-kV/500-MV A cable circuit. [20] investigated a critique of existing thermal models like IEC 60287 and Electra 143 for medium-voltage (MV) power cables in conduit installations, highlighting their oversimplifications. The authors employed laboratory measurements and finite element analysis (FEA) to investigate natural convection and air-gap thermal resistance. The authors found that the Raithby and Hollands Nusselt number correlation closely matched measured values, advocating for real-time installation environment modeling for more accurate cable ratings.

Numerous researchers have focused on the characteristics of the soil in which underground cables are buried, particularly in the case of non-homogeneous soil [10], these studies examined the non-uniform underground temperature distribution under different installation conditions and its impact on cable ampacity [21], to predict temperature distribution around existing underground cable systems, researchers conducted an experimental study on a scale model of a buried electric cable with different configurations and thermal properties of the mother soil and materials used to fill the trench [22]. In order to optimize the installation of buried electrical cables, the paper presented the issue of insufficient experimental data for electrical systems buried in non-homogeneous soils. The authors investigated this gap by creating a scale model to experimentally validate a numerical model and a simplified method they had previously proposed. The authors evaluated the thermal resistance of the soil. The authors found that the experimental data closely aligned with the numerical model [23]. The accurate calculation of the ampacity (maximum current-carrying capacity) of underground power cables is affected by soil moisture content [24]. To develop an improved thermal model for underground power cables, specifically in relation to cyclic ratings and thermal analysis, the authors conducted experimental work on eight different soil types. They determined the relationship between moisture content and specific heat with soil temperature and identified the temperature at which the formation of the dry zone occurs for each soil type [25]. The effect of the dry zone formation on the underground power cables capacities is investigated in [26].

Vollaro et al. [27] presented the limitations of the IEC 60287 standard, which often oversimplifies the thermal resistance calculations for underground electrical power cables by assuming homogeneous soil. The authors employed a control-volume formulation of the finite-difference method and conducted extensive numerical simulations. The authors derived a semi-empirical correlating equation with a 3.6 % standard deviation of error, proving to be more reliable and efficient than the existing IEC 60287 standard. Continuing this line of research, Maximov et al. [21] discussed the issue of inadequate models for calculating underground cable ampacity due to non-uniform soil temperatures. The author employed mathematical equations and iterative methods to develop a more realistic model. The results validated the proposed model as a more general formulation than existing ones, suggesting room for further improvement by considering environmental heating factors. Ocloñ, Pawel, et al. [10] presented an in-depth analysis of the shortcomings of prevalent thermal models such as IEC 60287, which frequently oversimplify underground power cable systems by assuming a homogeneous soil structure with constant thermal conductivity. To address this, the authors utilized the FEM to simulate a more complex, multilayered soil environment and different cable bedding conditions. The authors also incorporated various boundary conditions to enhance the realism of their simulations. The study concluded that the FEM simulations were in good agreement with ANSYS simulations, suggesting that their methodology provides a more accurate and comprehensive framework for assessing heat dissipation processes in underground power cables situated in thermal backfill and multilayered

soil.

Several techniques have been employed to optimize and study various aspects of underground power cable systems. In Ref. [28], the authors used a radial basis function neural network (RBNN) and a genetic algorithm (GA) to optimize the laying parameters of parabolic-type fluidized thermal backfill (FTB) trefoil buried cables, aiming to enhance the overall cable performance through proper geometric design. In Ref. [29], the authors introduced an adaptive optimized particle swarm optimization (PSO) algorithm combined with a back propagation neural network (LFVPSO-BPNN) and further enhanced it with GA. This solution was used to accurately predict the temperature and maximum current capacity of trench-laid cables. The authors found their combined approach to be reliable, accurate, and computationally efficient, closely aligning with COMSOL multiphysics simulations. Ref. [30] utilized a modified jaya algorithm and the FEM to minimize the total material costs of an underground power cable system while ensuring the cable conductor temperature did not exceed 65 °C. In Ref. [31], the authors employed the FEM and the PSO method to optimize material costs for a 400 kV cable line in a flat formation, aiming to ensure safe and reliable operation by keeping the cable core temperature below 90 °C. Finally, Ref. [32] discussed the complex issue of optimizing both material costs and electric-thermal performance in underground power cable systems (UPCS). The authors introduced a modified jaya algorithm (MJaya) and validated it against existing methods like PSO. They found that MJaya excelled in minimizing cable backfill costs and maximizing allowable electric current, outperforming PSO in these metrics.

Based on the related works discussed, this paper presents a novel approach for the thermal analysis of underground power cables. Application is carried out on 132 kV with a geometric cross-section area of 749.9 mm². The FEM is employed to calculate temperatures around cables, taking into account various soil types, cable installation configuration, and thermo physical properties. Subsequently, BPNN is applied for temperature estimation and prediction. The structure of the analysis conducted in this study is outlined as follows:

- (i) This paper presents an analysis of the one-term transient state, which represents the maximum load condition. The analysis examines this state under various environmental conditions.
- (ii) This study analyzes the impact of both homogeneous and non-homogeneous soils on cable heat dissipation and heat diffusion, with specific investigations into the influence of variations in soil composition and thermal properties, particularly soil thermal conductivity. The presence of dry zones and their effects are also examined. The scope of this research is extended to include the impact of two types of backfill materials, namely SCMB and FTB, on heat transfer around underground cables and how these materials influence heat dissipation and cable performance. Additionally, potential strategies to mitigate the challenges posed by variations in soil thermal properties on cable performance are explored.
- (iii) The study examines the effects of different geometry models for underground cables on heat dissipation and heat diffusion from the cable core to the surface ground, assessing the efficiency of various cable configurations such as flat and trefoil arrangements, and analyzing the influence of cable spacing, depth, and installation methods on heat transfer management.
- (iv) Given the complexities of the FEM, which demands the construction of a complex geometric model within a proficient working environment to obtain highly accurate results, this paper presents a novel approach. This innovative approach principally relies on an adaptive model powered by BPNN. The proposed BPNN model utilizes historical data to improve its predictions, enhancing its efficiency and accuracy. Moreover, the model undergoes a process of fine-tuning, where it adjusts and optimizes its internal parameters according to new incoming data. The objective of this fine-tuning process is to improve the model's

performance in predicting the temperature of underground power cables. This approach not only reduces the computational complexity involved in employing FEM alone but also leverages the power of machine learning to offer accurate and timely predictions. This fusion of traditional simulation methods with contemporary machine learning techniques signifies an important step towards smarter and more efficient systems for monitoring and managing underground power cables.

2. Materials and methods

2.1. Influential factors on cable performance

This paper focuses on the various factors that affect the performance of underground cable systems as illustrated in Fig. 1(a). These factors include the surrounding environment, which can consist of either single-layer and multi-layer soils that are composition-dependent (homogeneous and nonhomogeneous), as well as the cable geometry and configuration, insulation, and other features such as armor or shielding. The backfill material which are SCMB and FTB, cable dimensions, depth of burial (H), and distance between adjacent cables (S) also play a significant role in the thermal behavior and performance of underground cables. Furthermore, the temperature of the ground surface (T_{sg}) is an important factor in determining the heat transfer from an underground cable to the surrounding environment. Understanding these factors and their impact on cable performance is critical for designing and operating reliable and safe underground cable systems. Proper consideration of these factors can also help to maximize the capacity of the cable and minimize the risk of overheating, thereby extending the lifespan of the cable and reducing maintenance costs. Referring to Fig. 1(b), it can be observed that there are three types of temperature rating: Rated conductor temperature 90 °C, emergency rating 105 °C > 140 °C, and short circuit cable rating 250 °C. In this paper, the cable core maximum temperature is taken as 90 °C.

2.2. Geometric modeling and volumetric heat flux calculation in cable conductor

Fig. 2(a) shows the structure of the cable used as a case study; its rating is 132 kV. Its maximum conductor temperature in service is 90 °C, copper conductor and cross linked polyethylene (XLPE) insulation. Fig. 2(b) shows the internal and external thermal resistance and cable dimensions. Table 1 is divided into two sub-tables: (A) presents the cable layers and the materials used to manufacture the layers and (B) presents the thermal conductivity characteristics of the cable layers, soil, and backfill. Fig. 3 shows the environments surrounding the buried cable, which is single-layer homogeneous soil, multi-layer homogeneous or non-homogeneous soil, as well as the geometric dimensions of the surrounding environment. Homogeneous soil is uniform in its composition and properties throughout the entire region being considered. In the context of underground cable systems, this means that the soil surrounding the cables has constant properties (e.g., thermal conductivity, density, moisture content) throughout the entire length of the cable run. On the other hand, non-homogeneous soil is not uniform in its composition or properties. Instead, it has variations in properties within the region being considered. In the context of underground cable systems, this means that the soil surrounding the cables has different properties in different sections of the cable run. For example, the soil may have different thermal conductivities or moisture contents in different sections. This can impact the thermal behavior of the cable and must be taken into account when designing and operating underground cable systems. In both homogeneous and non-homogeneous soil conditions, multilayered soil consists of native soil (with a thermal conductivity of 1 W/mK), medium soil (with a thermal conductivity of 2.28 W/mK), and sand clay loam (with a thermal conductivity of 1.54 W/mK). Additionally, two types of backfill materials, namely SCMB and FTB, are

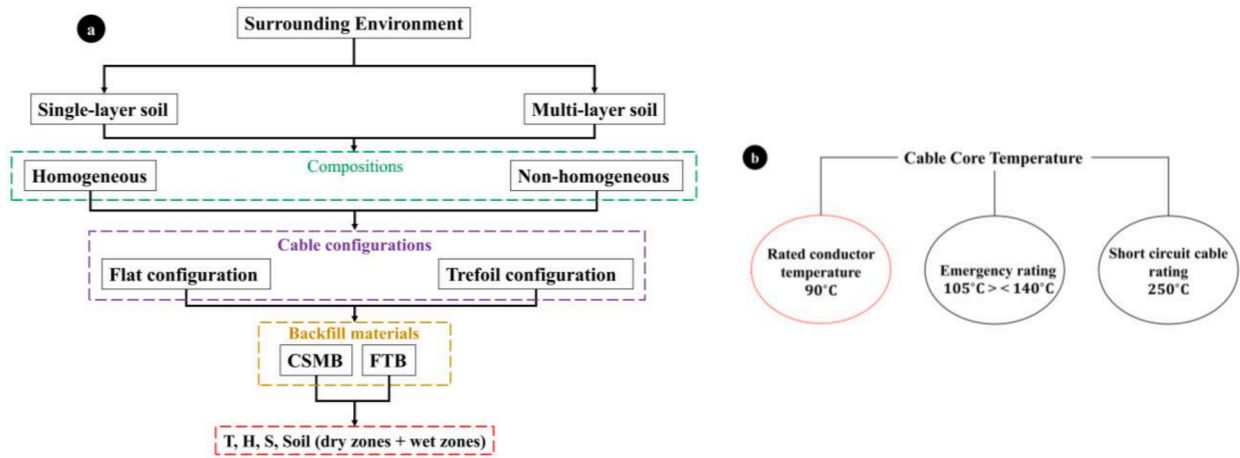


Fig. 1. Thermal considerations for underground cables, (a) role of surrounding environment, cable configuration, and backfill material, and (b) temperature ratings.

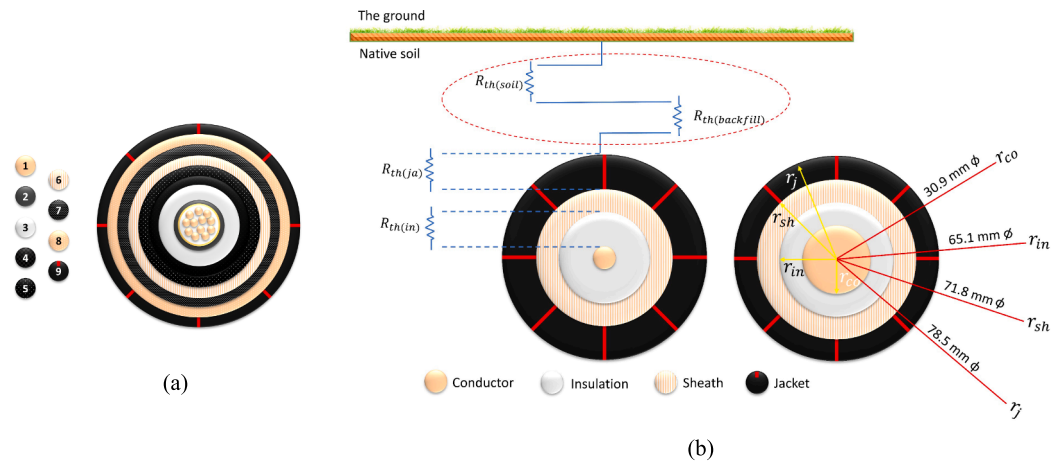


Fig. 2. Cable construction details, (a) cable layers, (b) power cable cross-section and thermal resistance.

Table 1

Cable layers material and physical properties.

(A)			(B)		
Number of layers	Elements	Material	Elements	Materials	Conductivity (W/mK) = $(1/\rho)$
1	Conductor	Copper	Conductor	Copper	386
2	Conductor screen	Semiconductor	Insulation	XLPE	0.2875
3	Insulation	XLPE	Jacket	HDPE	0.2875
4	Insulator screen	Semiconductor	Sheath	Copper	386
5	Swelling tape	Semi-conductive	Native soil	—	$1(k_{wet} + k_{dry})$
6	Wire screen	Copper	Medium soil	—	$2.28(k_{wet} + k_{dry})$
7	Swelling tape	Semi-conductive	Sand clay loam	—	$2(k_{wet} + k_{dry})$
8	Radial sealing	Aluminum	Backfill	SCMB	1
9	Jacket	HDPE	Backfill	FTB	1.54

employed. In non-homogeneous conditions, the temperature of the soil layers approaching the maximum allowable cable temperature results in the formation of dry zones around the power cables. These dry zones cause variations in the thermal conductivities of the soil layers surrounding the cables. The effects of these various factors on the thermal behavior of underground cable systems are analyzed to provide a better understanding of how the surrounding environment impacts the performance of buried cables [26,27].

The volumetric heat flux is a key parameter in this analysis, as it quantifies the rate of heat energy transfer per unit volume in the cable

conductor, as described by the following equation:

$$q_v(T_c) = \frac{W_c}{A} \quad (1)$$

where $A = \pi r_{co}^2 = 749.9 \text{ mm}^2$ is the geometric cross-section area of the conductor, and W_c is the cable heat losses (Joule losses) can be calculated through the equation below:

$$W_c = I^2 R_{ac} \quad (2)$$

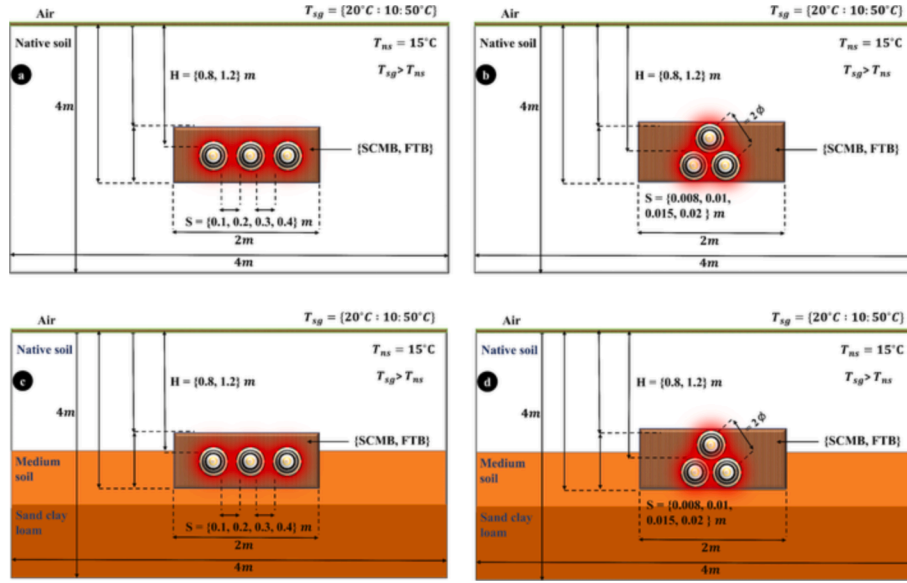


Fig. 3. The geometric model of cable burial, a) flat formation in homogeneous, b) trefoil formation in homogeneous, c) flat formation non-homogeneous, and d) trefoil formation in non-homogeneous.

where R_{ac} is AC conductor resistance (Ω/m), it is computed using the subsequent equation according to IEC 60287-1-2 [33]:

$$R_{ac} = R_{dc}(1 + \gamma_s + \gamma_p)$$

$$R_{dc} = R_{20}(1 + \alpha_{20}(T - 20)), \quad \text{for copper } \alpha_{20} = 3.93 \times 10^{-3} \quad (3)$$

$$R_{20} = \frac{\rho_{20}}{A}, \quad \text{for copper } \rho_{20} = 1.68 \times 10^{-8}$$

Given the temperature-dependent nature of the conductor resistance, which changes with temperature, R_{dc} (Ω/m) is the actual conductor resistance at temperature T . γ_s , γ_p are skin effect and proximity effect, respectively, according to IEC 60287-1-2 [33]:

$$\gamma_s = \frac{x_s^4}{192 + 0.8x_s^4} \quad (4)$$

where α_{20} is the temperature coefficient and T is the ambient temperature.

$$x_s^2 = \frac{8\pi f}{R_{dc}} 10^{-7} k_s, \quad k_s \text{ this value is given by the standard table} \quad (5)$$

$$\gamma_p = \frac{x_p^4}{192 + 0.8x_p^4} \left(\frac{d_c}{s} \right)^2 \left[0.312 \left(\frac{d_c}{s} \right)^2 + \frac{1.18}{\frac{x_p^4}{192 + 0.8x_p^4} + 0.27} \right] \quad (6)$$

$$x_p^2 = \frac{8\pi f}{R_{dc}} 10^{-7} k_p, k_p \quad (7)$$

this value is given by the standard table

where d_c is conductor diameter and s is the distance between the conductor axis x_s and x_p are argument of a Bessel function used to calculate skin effect and proximity effect.

2.3. Adaptive FEM-BPNN architecture

$$GDM = \begin{bmatrix} S_1 & S_2 \\ x_{min} & x_c \\ x_{max} & y_c \\ y_{min} & r \\ 0 & y_{max} \end{bmatrix} \quad (8)$$

For a rectangle $S_1 = 1$, $x_{min} = x_1$, $x_{max} = x_2$, $y_{min} = y_1$, $y_{max} = y_2$, and for a circle $S_2 = 2$, the center coordinates are (x_c, y_c) , and the radius is r .

This step is followed by the selection of the appropriate edge labels and face indices for utilization in the boundary equations. Subsequently, we generate a triangulated mesh, which acts as a discretized representation of the geometric domain of the problem; in essence, this constitutes a network or grid composed of small, interconnected elements. Once the mesh is established, we proceed to set the thermophysical properties, these include thermal conductivities, mass density, and specific heat for the various cable layers, soil layers, and backfill bedding.

The partial differential equation, known as the transient heat conduction equation, can be expressed as:

$$\nabla \cdot (k \nabla T) + q_v = \rho C_p \frac{\partial T}{\partial t} \quad (9)$$

In transient conditions, where temperature varies with time ($\partial T / \partial t \neq 0$), the heat conduction equation is:

$$\nabla \cdot (k \nabla T) = \frac{\partial}{\partial x} \left(k_{materials}(T) \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_{materials}(T) \frac{\partial T}{\partial y} \right) \quad (10)$$

For the conductor (internal heat source):

$$\frac{\partial}{\partial x} \left(k_{con}(T) \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_{con}(T) \frac{\partial T}{\partial y} \right) = \rho c \frac{\partial T}{\partial t} - q_v(T_c) \quad (11)$$

For the insulator, external layers, backfill, and soil:

$$\frac{\partial}{\partial x} \left(k_{ins, ex, bf, so}(T) \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_{ins, ex, bf, so}(T) \frac{\partial T}{\partial y} \right) = \rho c \frac{\partial T}{\partial t} \quad (12)$$

ρ is the density of the material, C_p is the specific heat capacity of the material, T is the temperature, and t is time. k_{con} , k_{ex} , k_{bf} , k_{so} , are the thermal conductivities of the conductor, external layers, backfill, and soil, respectively. ∇ represents the gradient operator, q_v is the volumetric heat generation rate due to the current passing through the cable. This is the heat source term that you can calculate based on the current and other cable properties.

Following this, we establish the boundary conditions and initial conditions. These correspond to the temperature at the point of contact with the soil surface and the initial soil temperature. Next, we designate the internal heat source, represented by the volumetric heat source (q_v) per unit volume of the cable conductor. Finally, after setting all con-

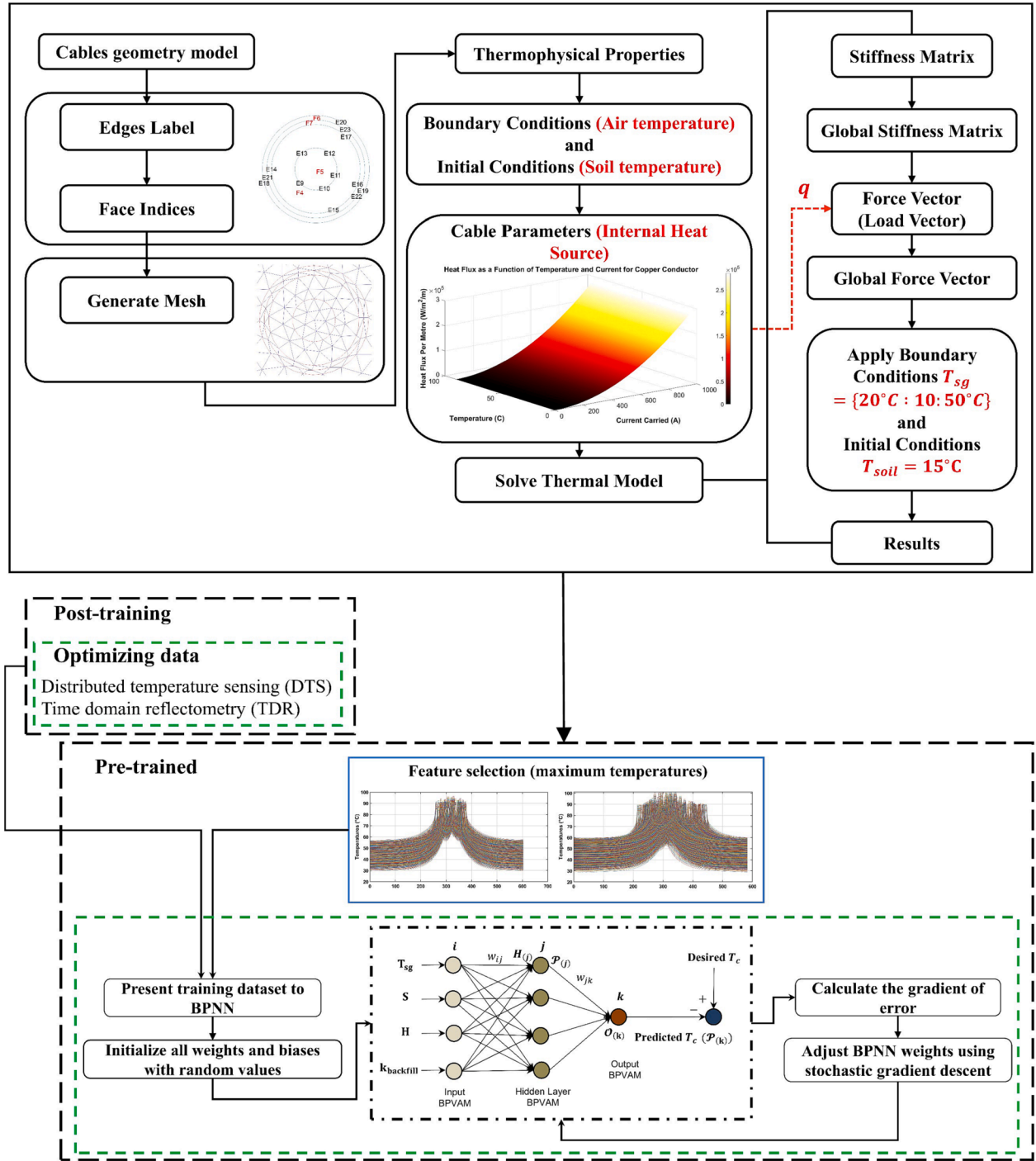


Fig. 4. FEM architecture for solving heat transfer in underground cables.

stants and variable values, we proceed to solve the thermal model. Our objective is to compute the temperature vector for each element in the homogeneous case and account for temperature-dependent thermal conductivity in non-homogeneous cases.

After calculation of the temperature vector for each element across all cases – including single-layer homogeneous soil, multi-layer homogeneous soil, and multi-layer non-homogeneous soil, the multidimensional aspects of cable geometry in these environments are considered. The transition then occurs to the feature selection algorithm [35], where the chosen feature is the maximum temperature. This process forms part of the preparation of a dataset for Training-BPNN. Each case has an input matrix that can be increased according to the case of the study, considering the

thermal conductivities of soil and other parameters that affect temperature distribution. BPNN utilizes stochastic gradient descent to adjust weights and minimize error, which is the difference between the target and predicted values. This process occurs after the initial feedforward operation ($X \times w_{ij} \rightarrow \sigma(x) = \frac{1}{1+e^{-x}} \times w_{jk} \rightarrow \sigma(x) = \frac{1}{1+e^{-x}} = p$, $e = t - p$), which employs weights and biases with randomly assigned values. Upon completion of the training process, the model becomes pretrained and ready for validation and testing. The learning behavior of the model can be optimized by upsampling a real dataset obtained from DTS or other sensors, specifically, for the dataset which suffers from inadequate knowledge that could provide better prediction performance.

3. Result and discussion

3.1. Cables in a single-layer homogenous soil

In this case, the underground cables are buried in a single-layer homogeneous soil, as depicted in Fig. 3(a) and (b). The burial configurations include both flat and trefoil, using two types of backfill materials – SCMB and FTB. The depth of the backfill is not constant; it varies with the depth of the cables. The homogeneous environment in this case is defined as native soil. The cables are buried at depths H ranging from 0.8 to 1.2 m, with the distance between cables S ranging from 0.1 to 0.4 m for flat and from 0.008 to 0.2 m for trefoil. The surface ground temperature T_{sg} is varied from 20 to 50 °C, while the soil temperature T_{soil} is maintained at a constant 15 °C. This section will discuss the effects of these factors on the thermal behavior and performance of the underground cable system.

Figs. 5 and 6 depict the temperature distributions surrounding the underground power cable system for Flat-SCMB and Flat-FTB, respectively, in single-layer homogeneous soil. Both configurations have cables at a depth of 80 cm and a distance of 40 cm between them. Figs. 5(a) and 6(a) display the maximum temperature curve on the X-axis for an environment width of -4 to 4 m, and the results show that the central conductor achieves a maximum temperature of 84.31 °C and 82.99 °C, respectively, due to mutual thermal effects. Figs. 5(b) and 6(b) show the maximum temperature curve on the Y-axis for an environment depth of 0 to 4.7 m, and the cable simulations in the Y-axis appear as a single cable in temperature distribution because the cable configuration is flat. Figs. 5(c) and 6(c) show 2D temperature distributions for the cables and surrounding environment. For the Fig. 6(c) the high thermal conductivity of the FTB material, which is typically around 1.54 W/mK, allows for efficient heat transfer between the cables and the surrounding soil, this can help to reduce the temperature rise of the cables and prevent overheating, which can help to ensure that the cables remain reliable and functional for longer periods. Furthermore, both figures show the direction of heat flux, which in this case would be from the cable towards the surrounding soil, since the cable temperature is higher than the temperature of the surrounding soil (14 °C). The surface ground temperature (20 °C) is between the cable and the surrounding soil temperature, so there would likely be a temperature gradient between the cable and the surface ground as well.

Figs. 7 and 8 both show temperature distributions surrounding underground power cable systems, with Fig. 7 depicting the Trefoil-SCMB configuration and Fig. 8 depicting the Trefoil-FTB configuration. In the flat configuration, higher mutual inductance and electromagnetic fields between the cables can result, which may lead to increased losses and interference. On the other hand, in the trefoil configuration, the three cables are evenly spaced and placed close together in an equilateral triangle formation, this arrangement helps to minimize the electromagnetic fields and mutual inductance between the cables while ensuring efficient use of space.

In the case of Trefoil-SCMB, the cables are at a depth of 80 cm, with a distance of 20 cm between them. This close distance results in mutual thermal effects, leading to increased cable temperatures and causing

thermal stress that can degrade the performance and lifespan of the cables. Fig. 7(a) shows the maximum temperature curve on the X-axis for an environment width of -4 to 4 m, with all three conductors achieving a maximum temperature of 86.10 °C due to mutual thermal effects. Fig. 7(b) shows the maximum temperature curve on the Y-axis for an environment depth of 0 to 4.7 m, with the cable simulation in the Y-axis appearing as double cables in temperature distribution because of the Trefoil configuration. Fig. 7(c) depicts 2D temperature distributions for the cables and surrounding environment, as well as the direction of heat flux, which would be from the cable towards the surrounding soil due to the cable temperature being higher than the temperature of the surrounding soil (14 °C).

In the case of Trefoil-FTB, the cables are also at a depth of 80 cm with a distance of 20 cm between them. Fig. 8(a) shows the maximum temperature curve on the X-axis for an environment width of -4 to 4 m, with all three conductors achieving a maximum temperature of 84.64 °C due to mutual thermal effects. The observed temperature decrease in the three conductors can be attributed to the high thermal conductivity of the FTB material, which facilitated efficient heat transfer between the cables and the surrounding soil. This allowed for the dissipation of heat outside the cables, thereby reducing the temperature of the conductors. Fig. 8(b) shows the maximum temperature curve on the Y-axis for an environment depth of 0 to 4.7 m, with the cable simulation in the Y-axis appearing as double cables in temperature distribution due to the Trefoil configuration. Fig. 8(c) depicts 2D temperature distributions for the cables and surrounding environment, as well as the direction of heat flux, which would also be from the cable towards the surrounding soil due to the cable temperature being higher than the temperature of the surrounding soil (14 °C). However, the high thermal conductivity of the FTB material, which is typically around 1.54 W/mK, allows for efficient heat transfer between the cables and the surrounding soil.

As observed in the figures that depicted the Y-axis temperature distribution, there is a fluctuation in the Y-axis perspective. This fluctuation is due to the method used to calculate the temperature distribution on the Y-axis, where the maximum value was taken from every 100 values. As a result, some points include values that represent lower temperatures.

3.2. Cables in a multi-layer homogeneous soil

In this case, the soil consists of three layers: native soil, medium soil, and sand clay loam (As depicted in Fig. 3(c) and (d)), each of these soil layers contributes to the overall behavior and characteristics of the soil surrounding the underground cables. The properties of the soil layers, including thermal conductivity, moisture content, and density, influence the heat dissipation and thermal performance of the cables. Each layer has its respective conductivity, as listed in Table 1.

Figs. 9–12 depict temperature distributions surrounding underground power cable systems in a multilayered homogeneous soil, comparing Flat-SCMB, Flat-FTB, Trefoil-SCMB, and Trefoil-FTB configurations. All configurations have cables at an 80 cm depth, with Flat-SCMB and Flat-FTB having a distance of 40 cm between cables, and Trefoil-SCMB and Trefoil-FTB having a distance of 20 cm between

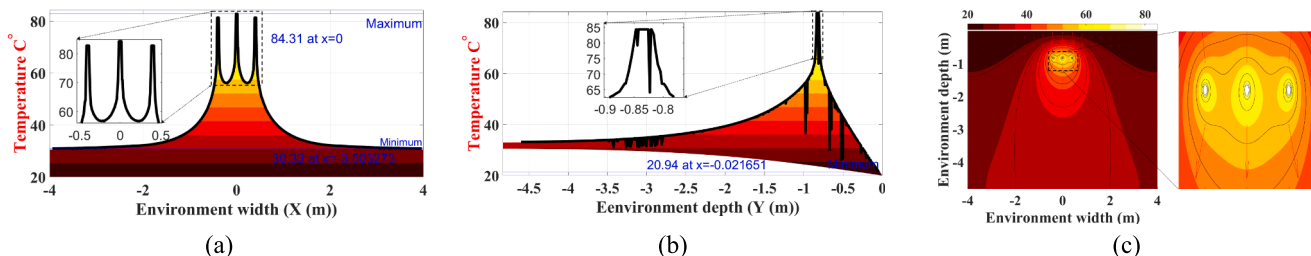


Fig. 5. Temperature distributions for Flat-SCMB in single-layer homogeneous soil.

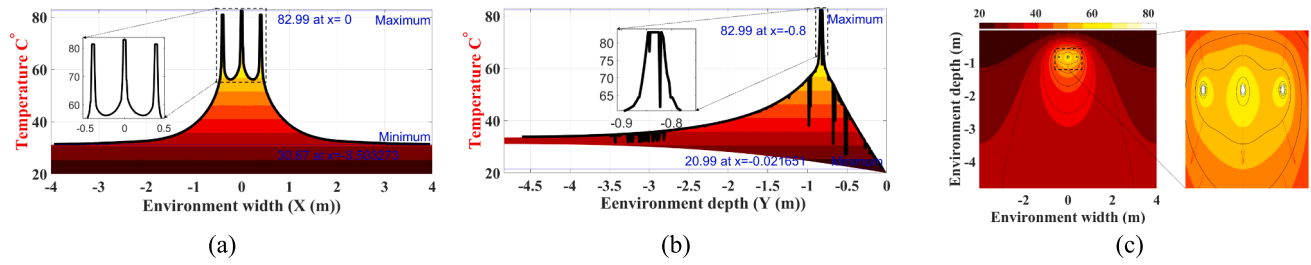


Fig. 6. Temperature distributions for Flat-FTB in single-layer homogeneous soil.

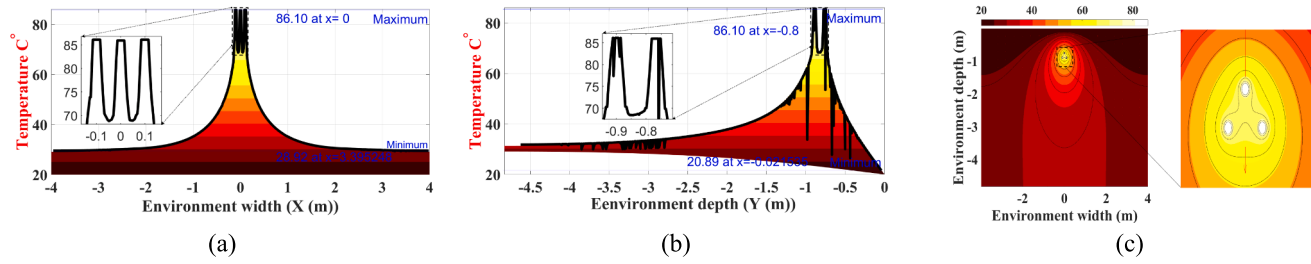


Fig. 7. Temperature distributions Trefoil-SCMB in single-layer homogeneous soil.

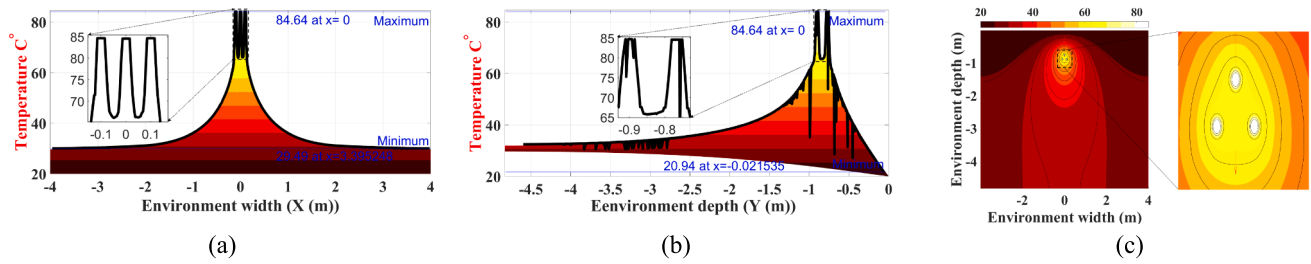


Fig. 8. Temperature distributions for Trefoil-FTB in single-layer homogeneous soil.

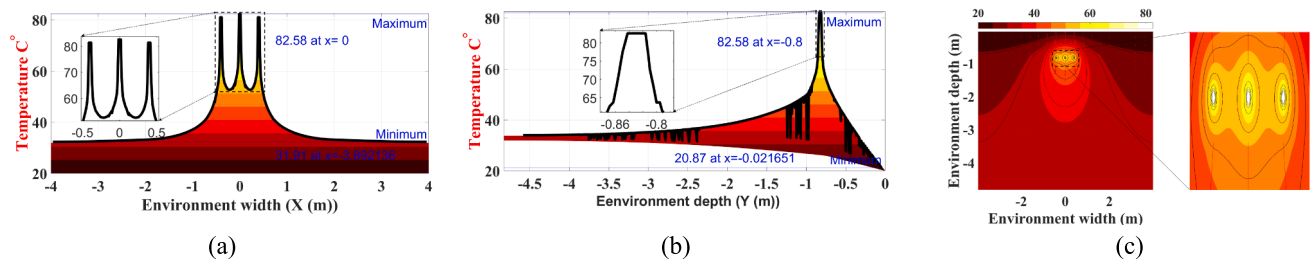


Fig. 9. Temperature distributions for Flat-SCMB in multi-layer homogeneous soil.

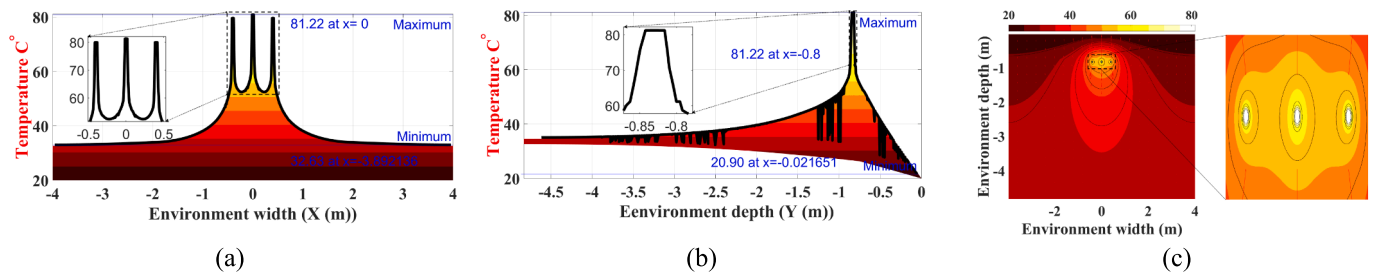


Fig. 10. Temperature distributions for Flat-FTB in multi-layer homogeneous soil.

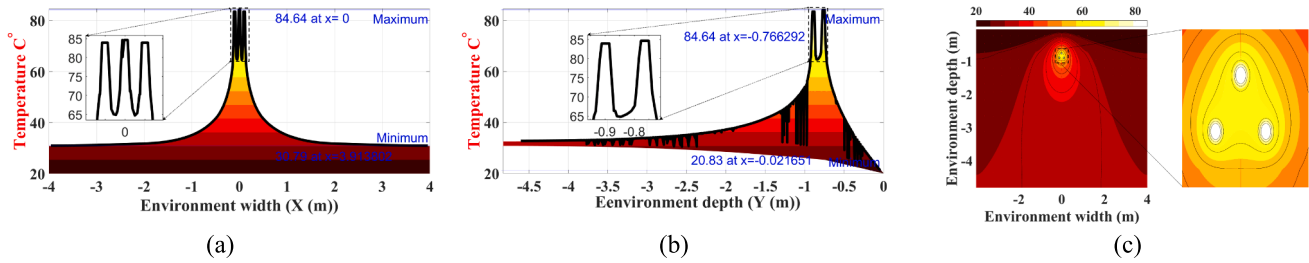


Fig. 11. Temperature distributions for Trefoil-SCMB in multi-layer homogeneous soil.

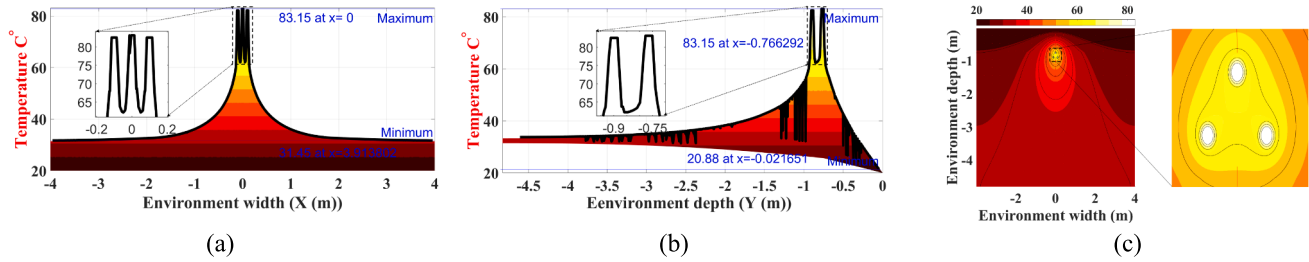


Fig. 12. Temperature distributions for Trefoil-FTB in multi-layer homogeneous soil.

cables. Figs. 9(a) and 10(a) show maximum temperatures of 82.58 °C and 81.22 °C, respectively, for the central conductor due to mutual thermal effects in Flat-SCMB and Flat-FTB configurations. Figs. 9(b) and 10(b) display the maximum temperature curve on the Y-axis for an environment depth of 0 to 4.7 m, with the cable simulations in the Y-axis appearing as a single cable in temperature distribution due to the flat configuration. The flat configuration allows for simplified analysis and easier interpretation of temperature patterns and their potential impact on cable performance. In contrast, Trefoil-SCMB and Trefoil-FTB configurations result in maximum temperatures of 84.64 °C (Fig. 11a) and 83.15 °C (Fig. 12a), respectively, for the three conductors. The Trefoil configuration helps to minimize electromagnetic fields and mutual inductance between the cables. For Flat-FTB (Fig. 10(c)) and Trefoil-FTB (Fig. 12(c)) configurations, the high thermal conductivity of the FTB material facilitates efficient heat transfer between cables and surrounding soil, reducing cable temperature rise and preventing overheating, ensuring cable reliability and functionality for longer periods. In all configurations, the heat flux direction is from the cable towards the surrounding soil due to the cable temperature being higher than the surrounding soil temperature, with a likely temperature gradient between the cable and surface ground, as shown in Figs. 9(c), 10(c), 11(c), and 12(c).

3.3. Cables in a multi-layer non-homogeneous soil

In the multi-layer soil cases, each layer consists of both dry and wet zones. Using the dry soil thermal conductivity k_{dry} value might not provide an accurate representation of the actual soil conditions surrounding the power cables. In real-world scenarios, the soil is rarely completely dry, and the presence of moisture can significantly affect the soils thermal properties. Moisture in the soil increases its thermal conductivity because water has a higher thermal conductivity than dry soil. This increase in thermal conductivity can impact heat dissipation and temperature distribution around the power cables, ultimately affecting the cables performance and ampacity. By using the moist soil thermal conductivity k_{wet} value, we account for real-world conditions where the soil contains some moisture. This approach provides a more accurate and relevant assessment of the cables thermal performance and its interaction with the surrounding soil (see Table 1).

Dry zones can form around power cables when the temperature of

the soil layers approaches the maximum allowable cable temperature. The current flowing through the cables produces heat, which is then transferred to the surrounding soil. In regions close to the cables, the soil temperature can rise significantly, resulting in the evaporation of moisture in the soil. As the moisture evaporates, the soil in the immediate vicinity of the cables can become drier compared to the surrounding soil. This creates areas with reduced moisture content, commonly known as dry zones. These dry zones lead to changes in the thermal conductivities of the soil layers surrounding the cables. This phenomenon occurs because the temperature increase affects the properties of the soil, including variations in thermal conductivity. In the case of multi-layered, non-homogeneous soil, we investigate the variability of thermal conductivity in both the insulator and the soil layers, as shown in Fig. 13. Based on these considerations, an analysis is performed to study the temperature distribution around the cable.

Figs. 14–20 shows the temperature distributions surrounding the underground power cable system in multi-layered, non-homogeneous soil for Flat-SCMB, Flat-FTB, Trefoil-SCMB, and Trefoil-FTB configurations, respectively.

Fig. 14–20(a and b) show the temperature distributions on the x-axis and y-axis, respectively. The central conductor achieved maximum temperatures of 83.65 °C in the Flat-SCMB configuration, 82.42 °C in the Flat-FTB configuration, 86.26 °C in the Trefoil-SCMB configuration, and 84.89 °C in the Trefoil-FTB configuration, and the minimum temperature achieved is 20.82 °C which is the upper of native soil.

Fig. 4 presents the architecture of the adaptive FEM-BPNN, which comprises both the FEM and BPNN components. Within the FEM segment, we commence with a geometric model of the cables and their adjacent environment as described in the following equation:

Figs. 14–20(c) shows the 2D temperature distributions, presenting isotherm regions. As observed in Fig. 3, the initial surface ground temperature is 20 °C, while the soil temperature is 14 °C. Notably, this soil temperature coincides with the initial temperature of the underground cables. Over time, the temperature difference results in the formation of a thermal gradient between the regions of high and low temperatures. This temperature variance leads to a significant change in thermal conductivity. As the cable's temperature elevates from 14 °C to 90 °C, each layer of the non-homogenous, multi-layered soil exhibits a variance in thermal conductivity, dependent on its component properties, as illustrated in Fig. 13(a). Upon closer inspection of Figs. 9–14(c) and

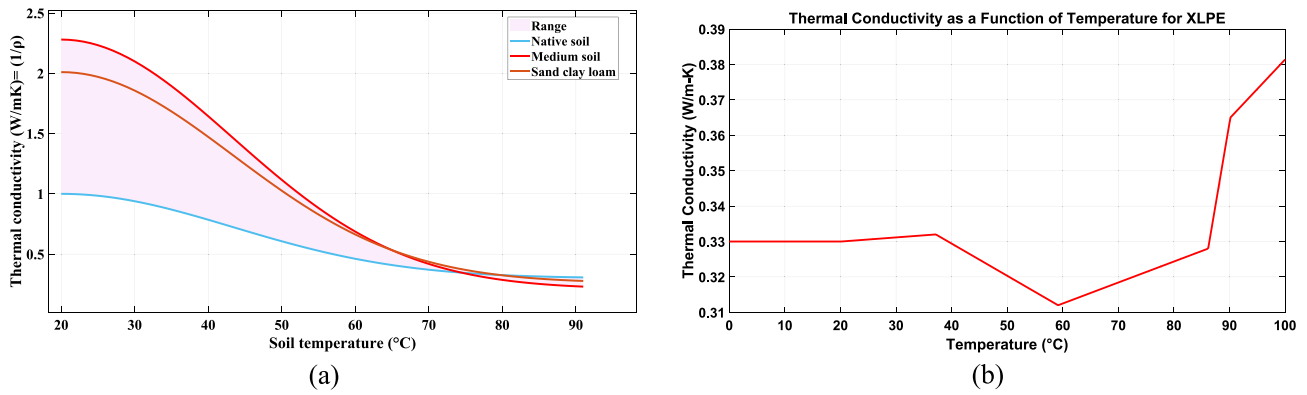


Fig. 13. The effect of an increase in cable core temperature which increases soil temperature on the conductivity of (a) soil layers [10], and (b) XLPE insulation [34].

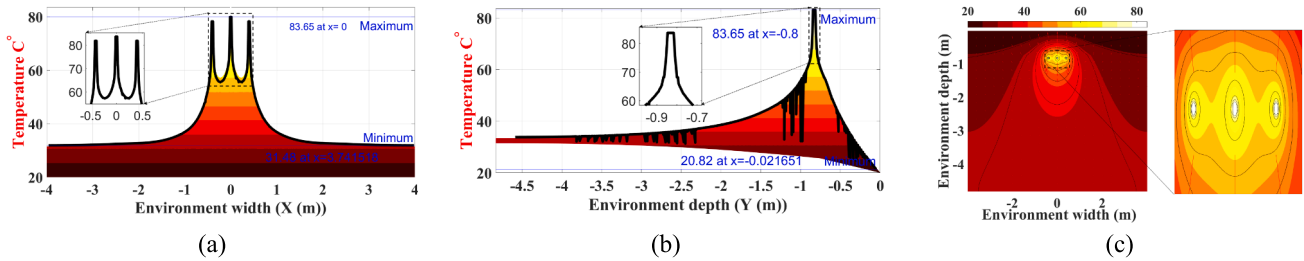


Fig. 14. Temperature distributions for Flat-SCMB in multi-layered, non-homogeneous soil.

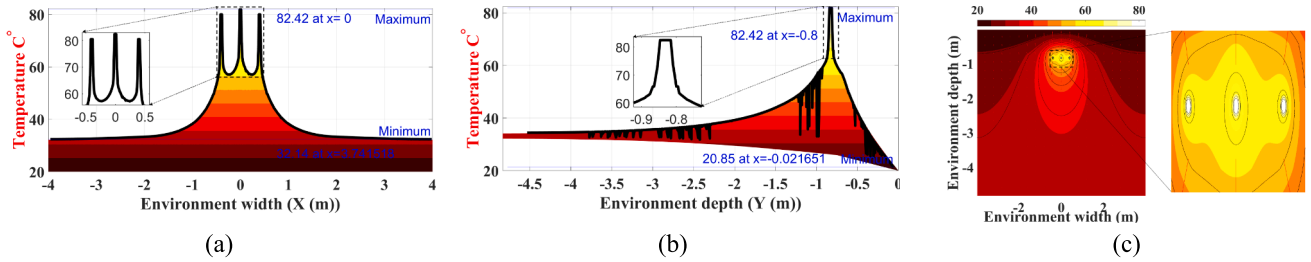


Fig. 15. Temperature distributions for Flat-FTB in multi-layered, non-homogeneous soil.

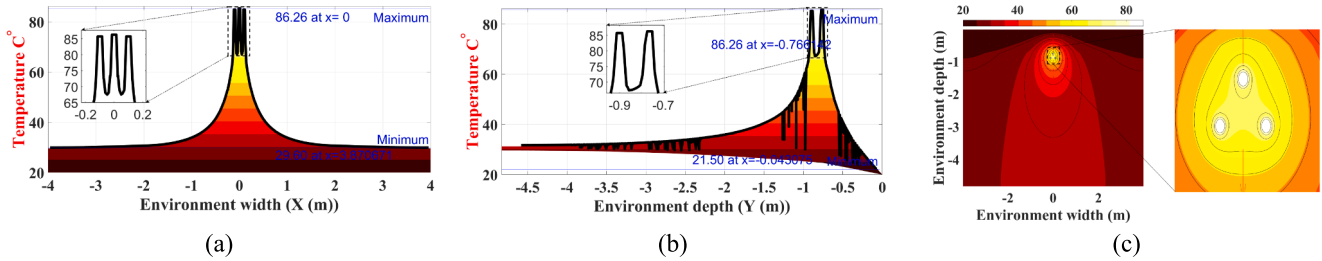


Fig. 16. Temperature distributions for Trefoil-SCMB in multi-layered, non-homogeneous soil.

Figs. 14–19(c), it becomes apparent that the yellow region, signifying approximately 60 °C, has migrated from the cable to its surroundings. Furthermore, in Figs. 11–16(c) and Figs. 16–20(c), a new region, denoted by a light yellow color, has formed as a result of a decrease in thermal conductivity.

3.4. Adaptive FEM-BPNN model for temperature prediction

In this section, three types of environments surrounding the cables were considered. The thermal behavior of these environments was

analyzed, and data were extracted based on this analysis to be tested in this section. In the three environments, the cables were configured in both flat and trefoil formations with maximum operation load 945 A; this was considered during the testing of the BPNN-FEM. A dataset matrix, with a size of 18600×8 , was extracted from the three environments, consisting of 6200 rows from each individual environment. The eight columns of the dataset matrix consist of surface ground temperature T_{sg} , distances between cables S , depth of cables H , and backfill conductivity $k_{backfill}$ along with an environment code. This code refers to the thermal conductivities of soil and other parameters that affect

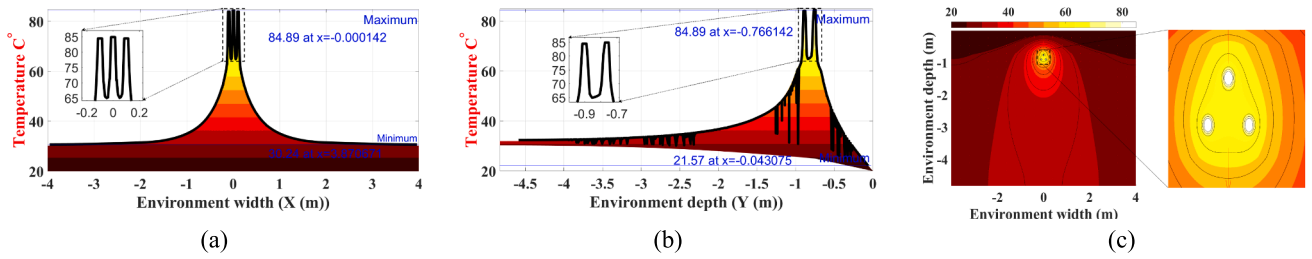


Fig. 17. Temperature distributions for Trefoil-FTB in multi-layered, non-homogeneous soil.

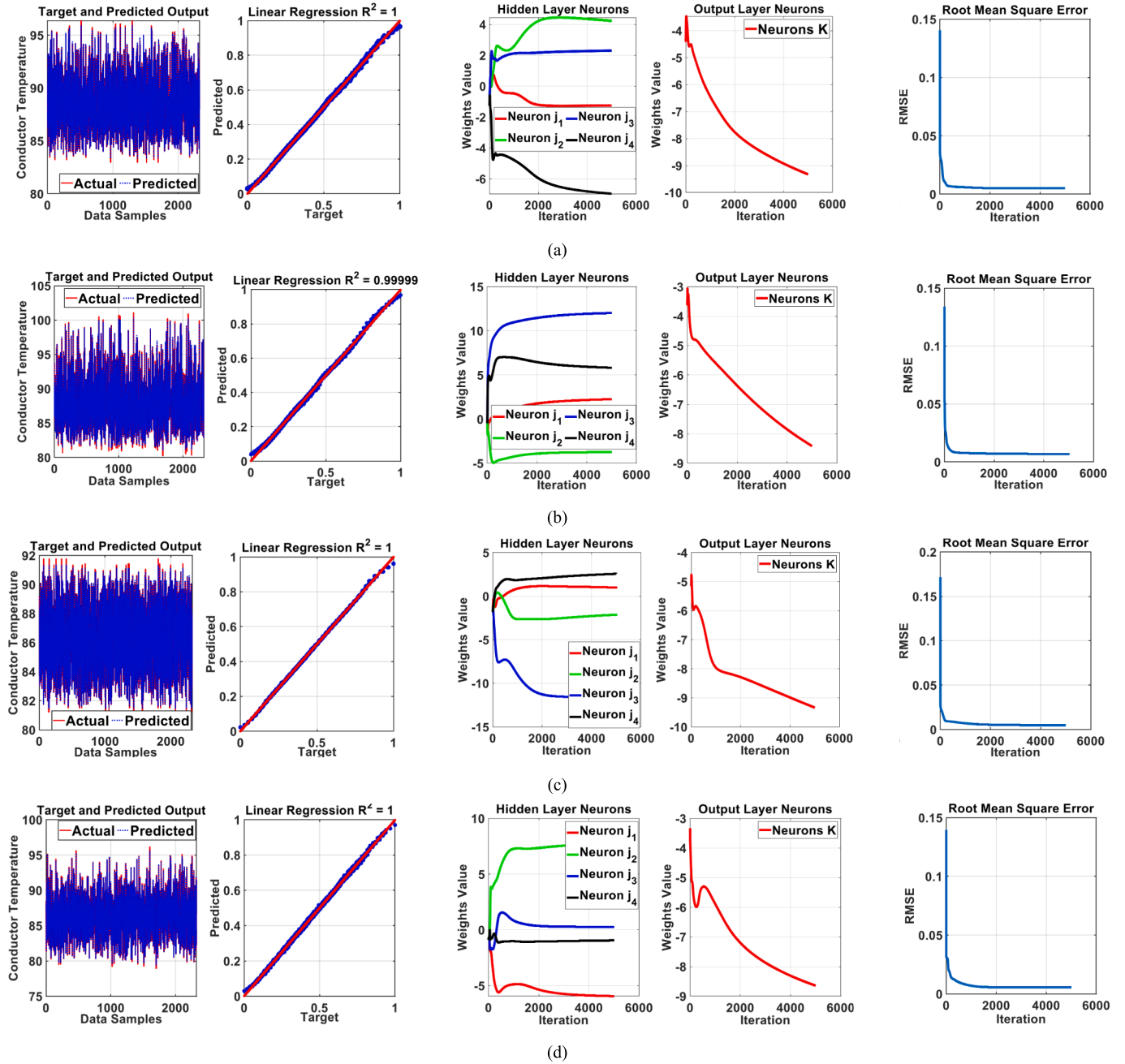


Fig. 18. BPNN training process for (a) single-layer homogeneous soil-Flat configuration, (b) single-layer homogeneous soil-Trefoil configuration, (c) multi-layer homogeneous soil-Flat configuration, (d) multi-layer homogeneous soil-Trefoil configuration, (e) multi-layer non-homogeneous soil-Flat configuration, and (f) multi-layer non-homogeneous soil-Trefoil configuration.

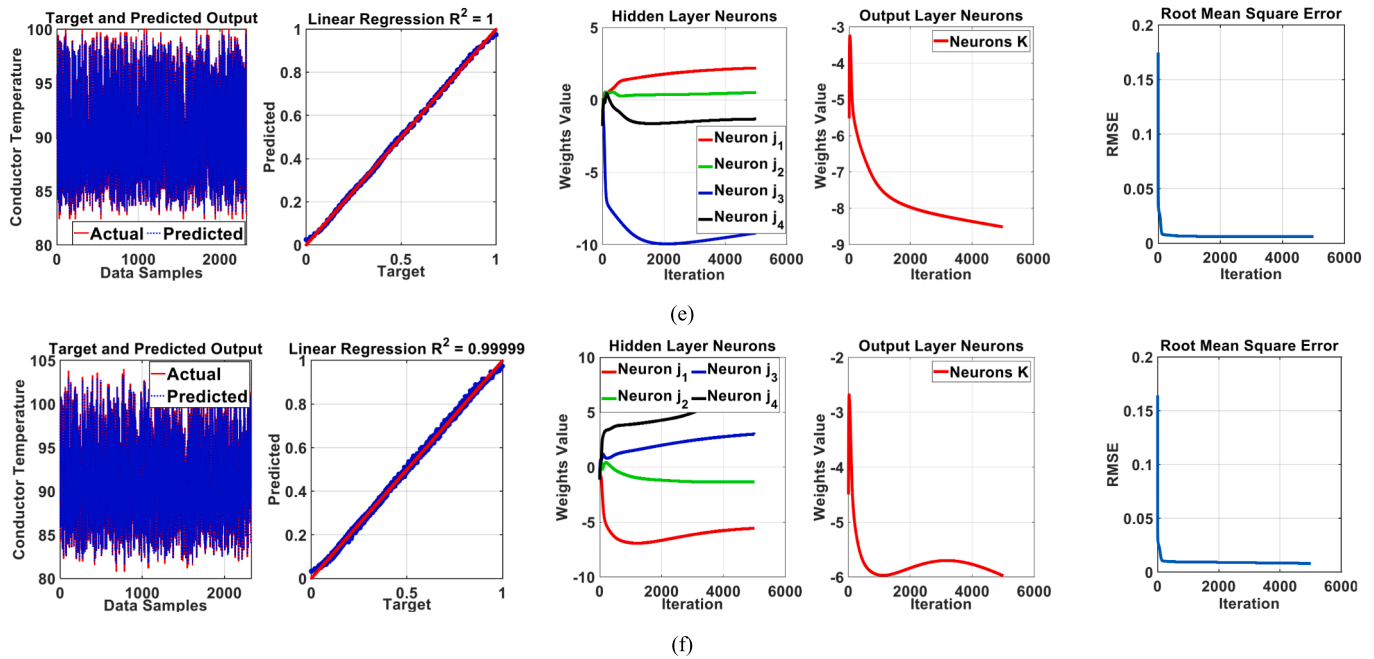


Fig. 18. (continued).

temperature distribution. The code consists of three binary numbers. The binary numbers for the single-layer homogenous soil environment are $[001]^T$ for flat and $[010]^T$ for trefoil, for the multi-layer homogeneous soil environment with flat and trefoil are $[011]^T$ and $[100]^T$, and those for the multi-layer non-homogeneous soil environment with flat and trefoil are $[101]^T$ and $[110]^T$. This code will increment the number of hidden neurons from 4 to 7.

In the preprocessing stage of BPNN-FEM, the dataset matrix is divided into inputs and outputs. The inputs include columns one through seven of the data matrices, whereas the outputs include the final column, comprising solely of maximum temperature distribution values. A subset, accounting for 75 % of the dataset matrix, is selected at random for the training process and 25 % for the testing process. These fractions are determined by multiplying the total number of rows in the data matrix by the prescribed percentage (0.75) and (0.25). A distinct permutation of row indices is generated to circumvent bias and ensure a randomized selection. Subsequent preprocessing involves the construction of a neural network model with seven hidden neurons, with a maximum iteration limit set at 5000. The activation functions for both the hidden and output layers are designated as sigmoid functions.

Fig. 18 illustrates the training process results of the FEM-BPNN model for various configurations. Fig. 18(a) presents the performance for the single-layer homogeneous soil-flat configuration. Similarly, Fig. 18(b) and (c) display the performance of the model for the single-layer homogeneous soil-trefoil and the multi-layer homogeneous soil-flat configurations respectively. The learning behavior of the model is also demonstrated for the multi-layer homogeneous soil-trefoil configuration in Fig. 18(d), for the multi-layer non-homogeneous soil-flat configuration in Fig. 18(e), and for the multi-layer non-homogeneous soil-trefoil configuration in Fig. 18(f). In all these scenarios, the model achieved a near-perfect or perfect linear regression coefficient of determination R^2 and maintained a low Root Mean Square Error (RMSE), ranging from 0.004 to 0.008. These results collectively confirm that the FEM-BPNN model adapts well to different environments and temperatures while ensuring its convergence. This adaptability and performance are indicative of an ideal model fit, with predictions aligning perfectly with the actual data. Furthermore, the learning behavior of the model is depicted through the alterations in weights for both hidden and output neurons, which essentially steer the learning

process. The diversity in the weights of the hidden layer indicates that the model operates with high accuracy and can process more data without weight conflicts.

The testing results corroborate the outcomes from the training and validation, demonstrating the effectiveness of the model during both the training and testing process. This is further evidenced by the RMSE values in the testing process: 0.005 for Fig. 19(a), 0.006 for Fig. 19(b), 0.004 for Fig. 19(c), 0.005 for Fig. 19(d), 0.005 for Fig. 19(e), and 0.008 for Fig. 19(f), as shown in Supplementary Table 1.

This research on FEM-BPNN for underground cable temperature prediction shows promise due to its novel approach (combining FEM accuracy with BPNN adaptability), improved accuracy (high R-squared, low RMSE), and ability to handle complex scenarios (different environments, non-homogeneous soils). This has the potential to optimize cable design and enhance power system operation.

3.5. Results comparison

Table 2 presents the results of conductor temperatures T_c for various underground power cable environments, configurations, backfill materials, and burial dimensions. It compares the temperatures for single-layer and multi-layer homogeneous soil, as well as flat and trefoil configurations, and two types of backfill materials: SCMB and FTB. The burial dimensions are represented by the distance between adjacent cables S and cable laying depth H . In the single-layer homogeneous soil, the highest T_c is observed for Trefoil-SCMB (86.10 °C), while the lowest is for Flat-FTB (82.99 °C). In multi-layer homogeneous soil, the highest T_c is seen in Trefoil-SCMB (84.64 °C), and the lowest in Flat-FTB (81.22 °C). Overall, the FTB material high thermal conductivity (1.54 W/mK) significantly contributes to heat dissipation in underground power cable systems, reducing cable temperature and extending their lifespan. This efficient heat transfer prevents overheating, which can cause thermal stress and degrade cable performance. The FTB configuration outperforms the SCMB configuration in heat dissipation and cable overheating prevention. In multi-layer homogeneous soil, consider factors like multilayered soil, FTB layer thermal conductivity, and potential dry zones for efficient underground power cable systems. Cable arrangement and soil thermal properties influence temperature distribution and performance. Flat configuration may offer better performance by the

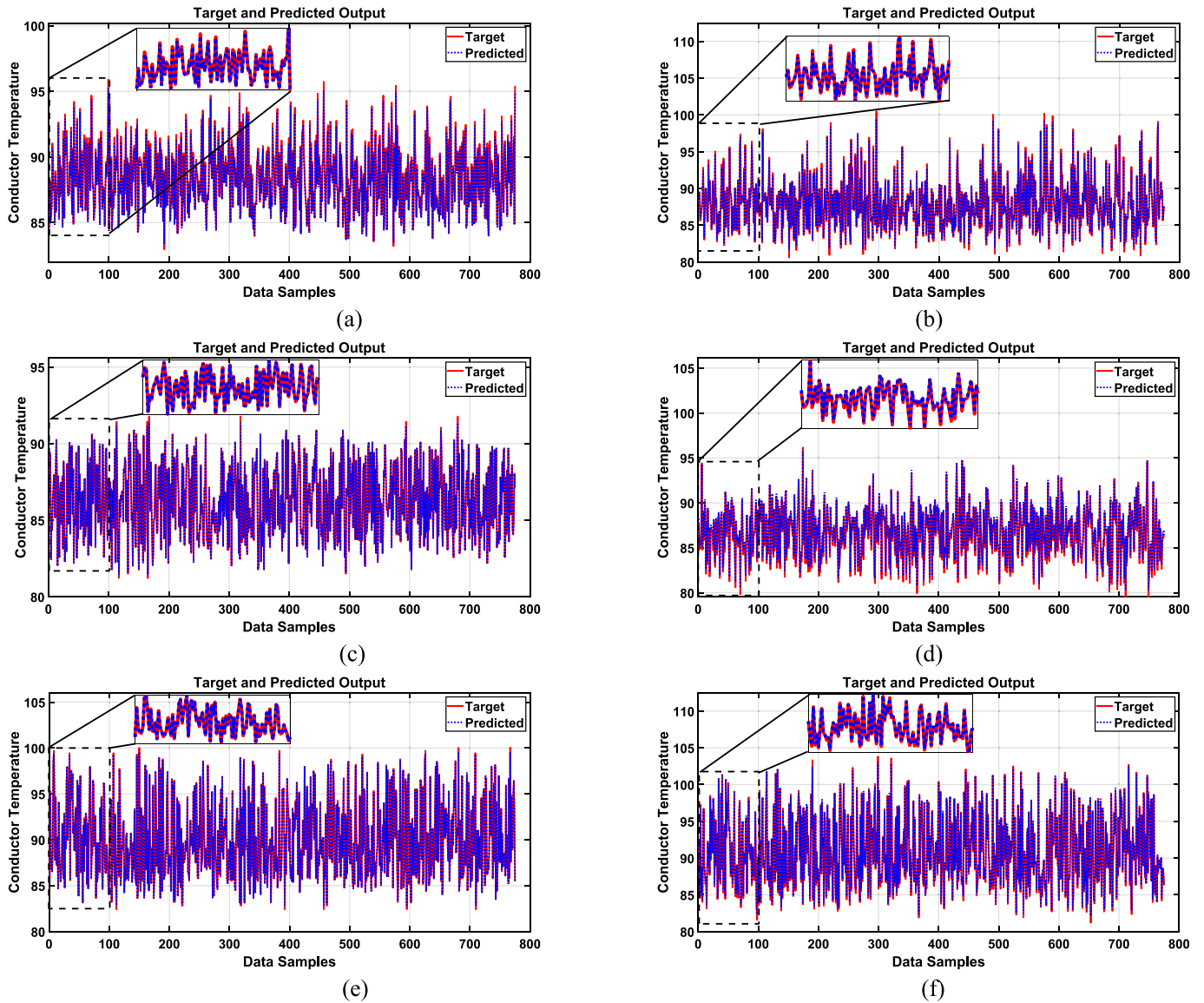


Fig. 19. BPNN testing process for (a) single-layer homogeneous soil-Flat configuration, (b) single-layer homogeneous soil-Trefoil configuration, (c) multi-layer homogeneous soil-Flat configuration, (d) multi-layer homogeneous soil-Trefoil configuration, (e) multi-layer non-homogeneous soil-Flat configuration, and (f) multi-layer non-homogeneous soil-Trefoil configuration.

ease of heat resulting from cable losses into the surrounding soil.

Table 3 compares the results of the study using the FEM with the Nexans and IEC [36] Standard-based values for cable, soil, electrical, and usage characteristics, considering a $I_{load, max}$ of 945 A and a T_{sg} of 20 °C. It presents two configurations, Flat and Trefoil, along with their respective soil conductivity values, burial dimensions, and T_c . The Nexans IEC Standard-based values indicate a T_c of 90 °C for both configurations, with a soil conductivity of 1.17 W/mK in the dry zone and burial dimensions of $S = 20$ and $H = 100$ cm. In contrast, the FEM results show lower conductor temperatures of 86.19 °C and 86.10 °C for the Flat and Trefoil configurations, respectively, with a soil conductivity of 1 W/mK and burial dimensions of $S = 20$ and $H = 80$ cm. This suggests that the FEM analysis in the study provides more efficient thermal performance in both configurations compared to the Nexans IEC Standard-based values. The last column in Table 2 shows the comparison between the results obtained in a single-layer and multi-layer homogeneous soil, respectively. The total conductor temperatures T_c we obtained are 480, which depend on the different case values for surface ground temperatures T_{sg} , distances between cables S , and the depth of the cables H . In this study, the multilayer soil achieved low conductor

temperatures, which depend on the high conductivities of the soil layers. However, in real-world situations, multi-layered soil is considered complex due to its varying properties and characteristics, which can significantly impact the performance of underground power cable systems.

Table 4 compares the temperature results for two different cases: multi-layer homogeneous soil and multi-layer non-homogeneous soil. For both cases, the temperature results are provided for Flat-SCMB and Flat-FTB configurations with the spacing S of 40 cm and burial depth H of 80 cm, similarly, the temperatures for Trefoil-SCMB and Trefoil-FTB with the spacing S of 20 cm and burial depth H of 80 cm, the temperatures for the central cable and the left and right cables are given for the conductor, insulator, sheath, and jacket. By comparing the temperature results of these two cases, it becomes clear that the rapid decrease in thermal conductivity of the soil layers influences the temperature gradient, taking this decrease into account results in more realistic outcomes.

To demonstrate the extensibility of our proposed approaches, we incorporated all environments. Fig. 20 presents an overall analysis of the training and testing results of the BPNN-FEM that encompasses all

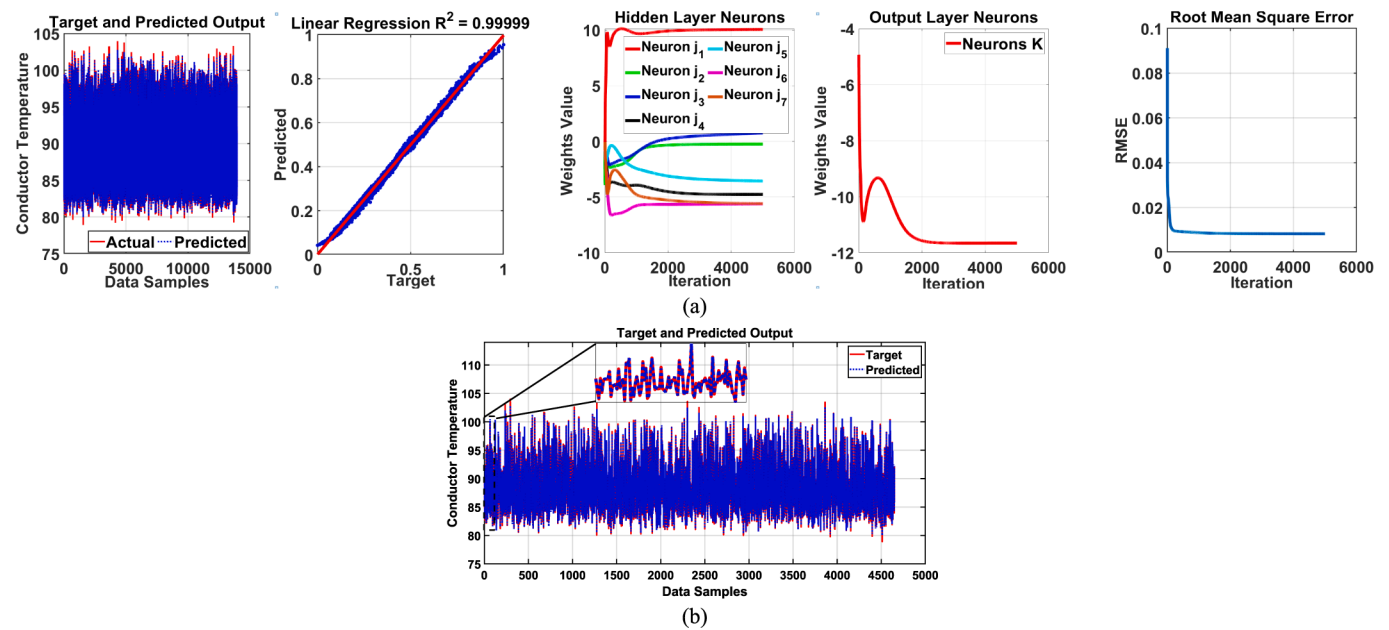


Fig. 20. Presents the results of the BPNN-FEM analysis, which encompasses all environments, (a) pertains to the training process, while part (b) illustrates the testing process.

Table 2
Conductor temperature under different conditions at $I_{load, max} = 945$ A and $20\text{ }^{\circ}\text{C } T_{sg}$.

Environments	Configurations	Backfill materials	Burial dimensions	Conductor temperature T_c	Overall results obtained
Single-layer homogeneous soil	Flat	SCMB	$S = 40, H = 80$ cm	$84.31\text{ }^{\circ}\text{C}$	
		FTB	$S = 40, H = 80$ cm	$82.99\text{ }^{\circ}\text{C}$	
	Trefoil	SCMB	$S = 20, H = 80$ cm	$86.10\text{ }^{\circ}\text{C}$	
		FTB	$S = 20, H = 80$ cm	$84.64\text{ }^{\circ}\text{C}$	
Multi-layer homogeneous soil	Flat	SCMB	$S = 40, H = 80$ cm	$82.58\text{ }^{\circ}\text{C}$	
		FTB	$S = 40, H = 80$ cm	$81.22\text{ }^{\circ}\text{C}$	
	Trefoil	SCMB	$S = 20, H = 80$ cm	$84.64\text{ }^{\circ}\text{C}$	
		FTB	$S = 20, H = 80$ cm	$83.15\text{ }^{\circ}\text{C}$	

Table 3
Comparing study results with Nexans IEC Standard-based for cable, soil, electrical, and usage characteristics at $I_{load, max} = 945$ A and $20\text{ }^{\circ}\text{C } T_{sg}$.

Methods	Configurations	Soil Conductivity (w/mk)= ($1/\rho$)	Burial dimensions	Conductor temperature T_c
IEC Nexans [35]	Flat	1.17 _{dry zone}	$S = 20, H = 100$ cm	$90\text{ }^{\circ}\text{C}$
	Trefoil	1.17 _{dry zone}	$S = 20, H = 100$ cm	$90\text{ }^{\circ}\text{C}$
FEM in this paper	Flat	1	$S = 20, H = 80$ cm	$86.19\text{ }^{\circ}\text{C}$
	Trefoil	1	$S = 20, H = 80$ cm	$86.10\text{ }^{\circ}\text{C}$

environments. In Fig. 20(a), the training results are noteworthy, achieving an impressive linear regression R^2 value of 0.99, paired with a low RMSE of 0.008. This demonstrates the BPNN-FEM high performance and accuracy during the training process. On the other hand, Fig. 20(b) Indicates the BPNN-FEM testing efficacy, it achieved a low RMSE of 0.008, further validating the robustness of the BPNN-FEM in effectively generalizing its learned patterns to unseen data. Finally, these findings confirm the uniform high performance of BPNN-FEM in various environments, consistently observed throughout both the training and testing process.

4. Conclusion

This study provided a novel approach for predicting underground cable temperature using an adaptive FEM-BPNN Model considering various soil compositions. An application using the proposed model and comprehensive thermal analysis for a 132 KV underground cable power system have been carried out. The analysis considered various factors

Table 4
Comparison of temperature results for different cases in multi-layer soil.

Cases	Configurations	Backfill materials	Burial dimensions	Temperatures °C		
Multi-layer homogeneous soil	Flat	SCMB	S = 40, H = 80 cm	Layers	Central cable	L&R cables
				Conductor	82.57	81.326
				Insulator	63.726	61.38
	Flat	FTB	S = 20, H = 80 cm	SheathJacket	61.56363.724	60.24361.378
				Conductor	81.213	79.92
				Insulator	61.183	58.741
	Trefoil	SCMB	S = 20, H = 80 cm	SheathJacket	58.85561.181	57.49658.739
				Conductor	84.634	83.919
				Insulator	67.6	66.21
	Trefoil	FTB	S = 20, H = 80 cm	SheathJacket	65.55167.598	64.98564.985
				Conductor	83.144	82.543
				Insulator	64.8	63.614
				SheathJacket	62.5564.798	62.2762.27
Multi-layers non-homogeneous soil	Flat	SCMB	S = 40, H = 80 cm	Layers	Central cable	L&R cables
				Conductor	83.643	81.804
				Insulator	68.159	64.797
	Flat	FTB	S = 20, H = 80 cm	SheathJacket	66.1268.157	63.63264.795
				Conductor	82.415	80.521
				Insulator	65.926	62.455
	Trefoil	SCMB	S = 20, H = 80 cm	SheathJacket	63.73365.924	61.17462.453
				Conductor	86.248	85.713
				Insulator	69.799	68.712
	Trefoil	FTB	S = 20, H = 80 cm	SheathJacket	67.88969.798	67.567.5
				Conductor	84.883	84.446
				Insulator	67.153	66.242
				SheathJacket	65.05867.151	64.91264.912

that affect the performance of the cable system, such as soil types and compositions, cable installation configuration, and thermophysical properties. Given their significant impact, a precise examination of these elements was essential. Soil types could be either single-layer or multi-layer, and their compositions could be homogeneous or non-homogeneous. cable installation configuration pertained to the depth of the cable burial and the distances between cables. Thermophysical properties involved variables such as thermal conductivities. The study goes more in-depth and includes the formation of dry zones and their effect on heat distribution around cables.

The results from the study clearly illustrated the impact of various factors on the thermal performance of underground power cables. In single-layer homogeneous soil, the highest conductor temperature T_c recorded was 86.10 °C for the Trefoil-SCMB configuration, while the lowest was 82.99 °C for the Flat-FTB configuration. In contrast, in multi-layer homogeneous soil, the highest T_c was 84.64 °C in the Trefoil-SCMB configuration, and the lowest was 81.22 °C in the Flat-FTB configuration. Moreover, the study conducted a comparison between multi-layer homogeneous and non-homogeneous soils. The findings underscored the significance of considering variable thermal conductivities. A notable decrease in thermal conductivity in non-homogeneous soil layers influenced the temperature gradient, leading to an increase in conductor temperature.

Furthermore, the study presented a novel approach based on FEM, referred to as BPNN-FEM. This model leverages historical FEM data to enhance its predictive power, thereby increasing efficiency and accuracy. Notably, the BPNN-FEM model is adaptable and refines its internal parameters in response to new data. BPNN achieved an impressive R^2 value of 0.99 and a low RMSE of 0.008 during both training and testing

processes.

The findings from this analysis significantly contribute to understanding and improving the performance and reliability of underground cable power systems. Another key point to note is that in real-world situations, multi-layered soil is complex, with various properties that can significantly impact the performance of these systems. Therefore, for a more accurate assessment, it is essential to consider the changing characteristics of soil layers and their potential effects on cable performance.

CRedit authorship contribution statement

Abdullah Ahmed Al-Dulaimi: Writing – original draft, Writing – review & editing, Methodology. **Muhammet Tahir Guneser:** Conceptualization, Methodology, Supervision. **Alaa Ali Hameed:** Methodology, Supervision. **Fausto Pedro García Márquez:** Formal analysis. **Osama E. Gouda:** Formal analysis, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The supplementary information presents the results of maximum temperature calculations for load currents I_{load} ranging from {805 A:945 A}. This analysis includes the first two scenarios: cables in a single-layer homogeneous soil and cables in a multi-layer homogeneous soil. Additionally, Supplementary Fig. 5 displays temperature calculation results based FEM for all considered cases, these are subsequently preprocessed to extract the maximum values, which are then used as targets for the BPNN. Furthermore, Supplementary Fig. 6 offers a detailed view of Fig. 18 from the main

paper, focusing on a sub-range for Fig. 6(a)–(f) within the range of {1:500} and for Fig. 6(g) within the range of {1:600}. Finally, we provide specifications of the cable used in our study in Supplementary Fig. 7. Supplementary Table 1 presents the RMSE results of the testing process for each case using the BPNN-FEM. The final row of the table provides the RMSE results for the testing process, which is based on training with the complete dataset from all cases.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jestch.2024.101658>.

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